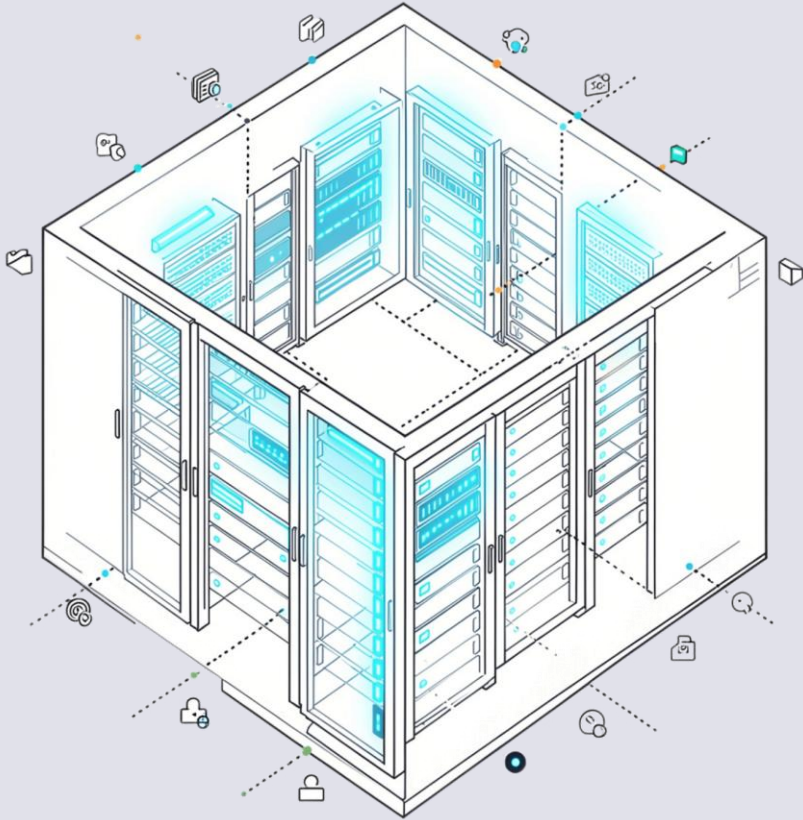




Module 2.2 – Data Warehouse and OLAP

Online Analytical Processing & Knowledge Discovery

Why Do We Need OLAP?

Real-world scenario: Flipkart's CEO needs instant answers to strategic questions — but the operational database can't keep up.

- Which products sold most during Diwali? Which city generated highest revenue?
- Which category is growing fastest? How have sales changed over 5 years?
- Operational databases cannot answer these questions instantly
- **Solution:** OLAP enables fast, multidimensional analysis of historical data



Key Insight: "OLTP runs the business. OLAP helps *improve* the business."



Classroom Question

Why can't Flipkart's operational database — which processes daily transactions — answer strategic questions about sales trends instantly?



OLAP?

Here **Online** means

"The user gets answers immediately while interacting with the system." No waiting for reports.

"What does **Analytical** mean?"

Instead of entering new data, we analyze existing historical data.

Processing Means

High-speed computation. Thousands or millions of records are processed in seconds.

Example:

"If Amazon stores 15 years of sales data, what questions can OLAP answer?"

Highest-selling category? Most profitable city? Seasonal sales? Customer spending pattern?

OLAP is not for storing data. OLAP is for analyzing data."

What is OLAP?

Analogy: Google Maps stores millions of roads. OLAP is like asking "Show all traffic congestion in Mumbai during office hours" — instantly analyzing massive datasets.

Enables interactive exploration across **time, geography, product, and customer** dimensions.

Online

Real-time query response — no waiting for batch reports

Analytical

Complex multi-dimensional analysis across dimensions

Processing

High-performance data computation at scale

🎯 **Classroom Question:** If you were analyzing Amazon's sales data across 50 product categories, 500 cities, and 10 years of history — how many different questions could you ask OLAP?

Benefits of OLAP



Faster Reporting

Generate complex reports in seconds instead of hours



Sales Forecasting

Predict future demand based on historical patterns



Better Decisions

Data-driven insights for strategic planning



Trend Analysis

Identify patterns across time, geography, and product categories



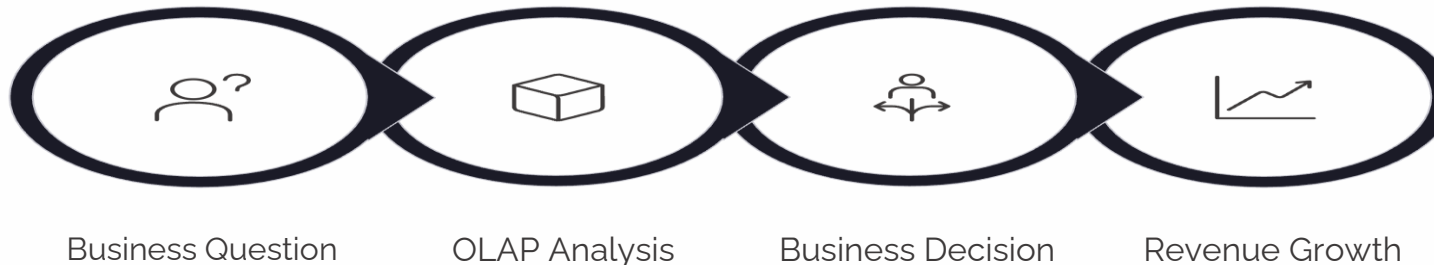
Interactive Dashboards

Explore data dynamically without IT intervention



Data Summarization

Aggregate millions of transactions into actionable insights

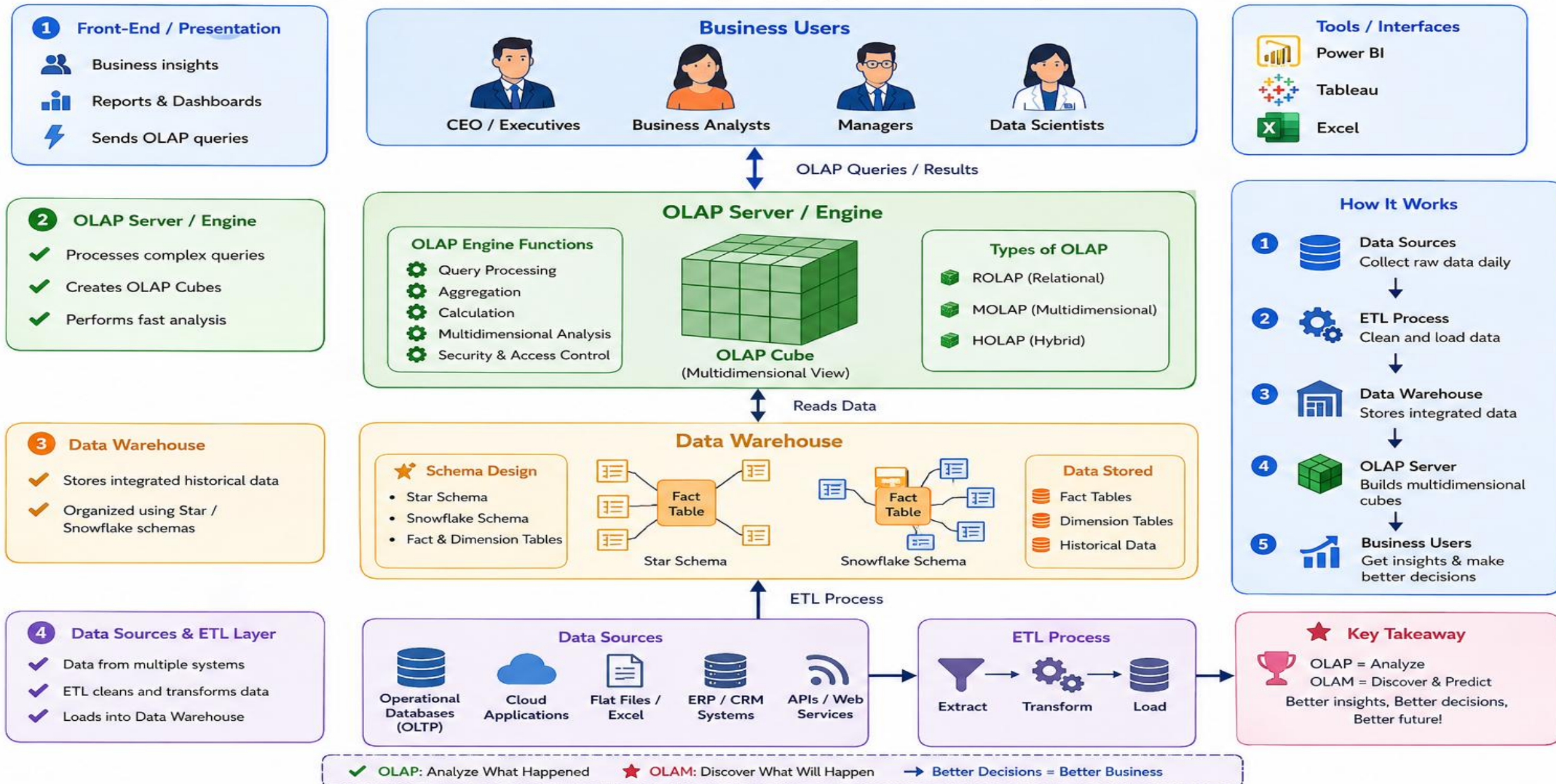


Flipkart example: "Electronics sales increase 40% during Diwali" → Marketing launches targeted campaign → 25% additional revenue generated.

🎓 **Classroom Question:** Which benefit of OLAP would be most valuable to Netflix: faster reporting, trend analysis, or sales forecasting? Why?

OLAP Architecture – Technical Block Diagram

OLAP sits on top of the Data Warehouse and provides fast, multidimensional analysis for business users.



OLTP vs OLAP: Real-Life Example at Flipkart

⚡ OLTP — Operational Database

- Records transaction in real-time
- Updates inventory (reduces stock by 1)
- Processes payment and generates invoice
- Sends order confirmation to customer

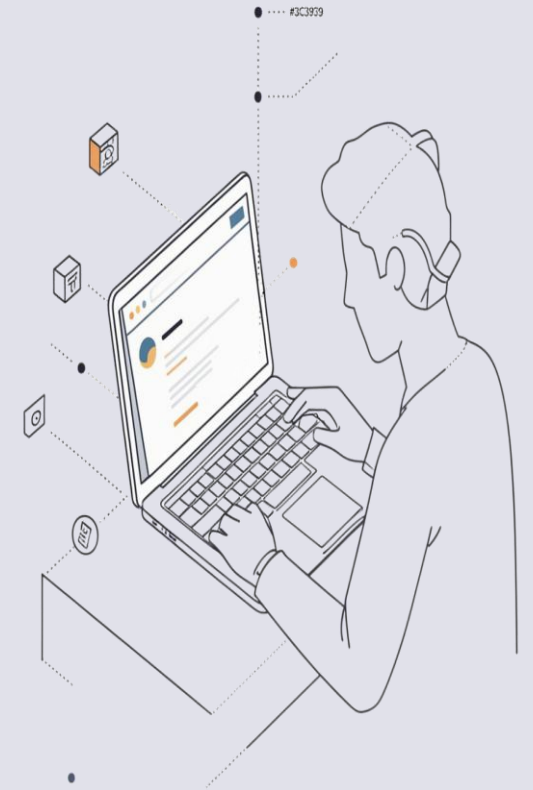
Focus: Immediate, accurate transaction processing

🏠 OLAP — Data Warehouse

- Aggregates monthly laptop sales across all cities
- Identifies top-selling brands and models
- Determines highest revenue city and customer segment
- Analyzes yearly trends and seasonal patterns

Focus: Strategic insights from historical data

🎯 **Classroom Question:** If Flipkart's OLTP system processes 10,000 transactions per second, how would running complex analytical queries on it affect the customer checkout experience?



OLTP vs OLAP: Comparison

Aspect	OLTP	OLAP
Purpose	Daily operations	Strategic analysis
Users	Clerks, operators	Managers, analysts
Data	Current, detailed	Historical, summarized
Operations	Insert, update, delete	Read, analyze, aggregate
Query Complexity	Simple, fast	Complex, multidimensional
Database Size	Small to medium	Very large (terabytes)
Examples	ATM, online shopping, railway reservation	Sales dashboard, market analysis, hospital reports

❏ **Key Principle:** OLTP supports daily business operations, while OLAP supports strategic decision making.

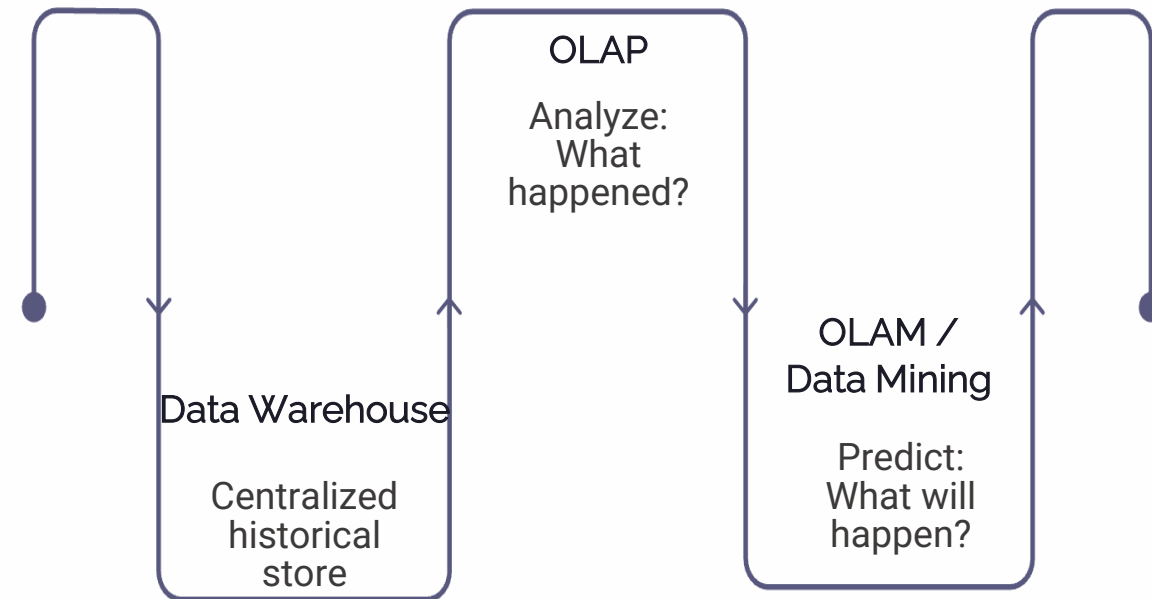
🔗 **Classroom Question:** Why would using OLTP (operational database) for strategic analysis slow down both the business operations and the analysis itself?

Introduction to OLAM

Knowing *what happened* is useful... But can we **predict what will happen next**?

- **OLAP limitation:** Answers "What happened?" but cannot predict future or discover hidden patterns
- **Solution:** Online Analytical Mining (OLAM)
- **Definition:** OLAM combines OLAP (analysis) + Data Mining (pattern discovery) for knowledge discovery
- **Real-world example:** Netflix doesn't just analyze what users watched (OLAP) — it *predicts* what they'll watch next (OLAM)
- OLAM enables organizations to move from **reactive to proactive** decision-making

🎯 **Classroom Question:** How would Amazon use OLAM to improve its recommendation system beyond just analyzing past purchases?



From OLAP to OLAM: The Evolution

OLAP Answers

- "What happened?" — Historical facts and trends
- "How much?" — Quantitative analysis
- "Which?" — Comparisons and rankings

OLAM Answers

- "What patterns exist?" — Hidden relationships in data
- "What is likely to happen?" — Predictive insights
- "What should we do?" — Actionable recommendations

Amazon Recommendation System — OLAP vs OLAM in Action

OLAP

"Customers bought 50,000 laptops last month."

OLAM

"Customers who buy laptops also buy laptop bags (78% correlation). **Recommend bags to laptop buyers.**"

🔗 **Classroom Question:** How would a hospital use OLAM to improve patient care beyond just analyzing historical patient records?

OLAM Architecture: Layered Design

01

Operational Databases

Daily transaction data — sales, inventory, customer records

04

OLAP + Mining Engine

Multidimensional queries + pattern discovery algorithms

02

ETL Layer

Extract, Transform, Load data from multiple sources

03

Data Warehouse

Centralized repository of cleaned, integrated historical data

05

Knowledge Base & Users

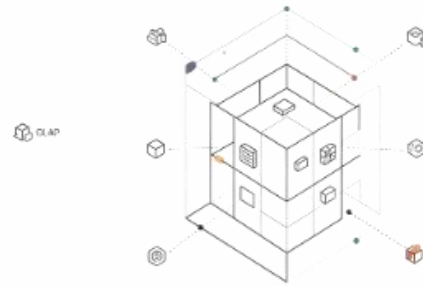
Insights delivered via dashboards and reports to decision-makers

Components of OLAM Architecture



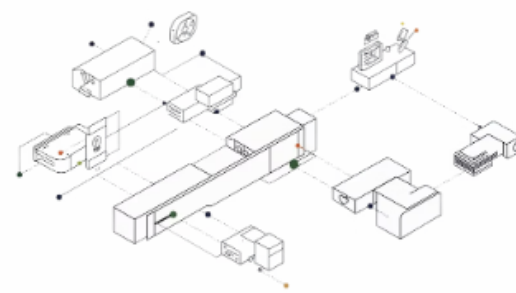
Data Warehouse

Stores 5–10 years of historical data in star/snowflake schema. Optimized for read-heavy analytical queries.



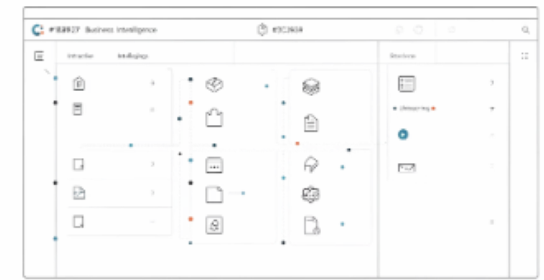
OLAP Server

Enables roll-up, drill-down, slice, dice, pivot. Returns results in milliseconds for interactive exploration.



Mining Engine

Applies classification, clustering, association rules, regression. Discovers patterns humans cannot see manually.



Knowledge Base & Visualization

Stores discovered rules for reuse. Presents insights through interactive dashboards and reports.

- ✔ **Netflix Result:** OLAM on 10 years of viewing history from 200M users → "Users who watched Stranger Things also watched The Crown (correlation: 0.85)" → Personalized recommendations increase watch time by **30%**.

OLAP vs OLAM: Complete Comparison

Aspect	OLAP	OLAM
Purpose	Historical analysis	Pattern discovery & prediction
Output	Reports, dashboards, summaries	Patterns, rules, predictions
Techniques	Aggregation, drill-down, pivot	Classification, clustering, association rules, regression
Questions Answered	What happened? How much? Which?	What patterns exist? What will happen? What should we do?
Flipkart Example	"Electronics sales grew 40% in Diwali"	"Customers aged 20–30 buy smartphones during Diwali; recommend accessories"

❏ **The bottom line:** OLAP tells us *what happened*. OLAM tells us *why it happened* and *what may happen next*.

🖋️ **Classroom Question:** If you were a data scientist at Swiggy, how would you use OLAP and OLAM *differently* to improve food delivery?